

GENERAL SPECIFICATION G5.50

Electrical Systems Analysis

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G5.50 ELECTRICAL SYSTEMS ANALYSIS

G5.50.1 General

This Specification details the minimum requirements for the Electrical Systems Analysis. This document shall be read in conjunction with all General Specifications G5, Particular Specifications P5, and Drawings.

(a) References

The following is a list of standards which may be referenced in this Section:

- (i) International Electro-technical Commission (IEC): 60909, Short-Circuit Calculations in Three-Phase A.C. Systems;
- (ii) General Specification G5.2- Standards, Code and Regulations;
- (iii) Occupational Safety and Health Standards (OSHA): 29 CFR, Part 1910 Subpart S, Electrical; and
- (iv) SPRING Singapore
 - 1) CP 5, Code of practice for electrical installations;
 - 2) SS 551, Code of practice for earthing;
 - 3) SS 555, Code of Practice for Lightning Protection; and
 - 4) Electricity Act (CHAPTER 89A).

(b) Submittals

- (i) Shop Drawings, Samples and Technical Data: As a minimum, submit the following Electrical System Analysis study and reports in accordance with General Specification G1 – Drawings, Documentation and Recording of Works:
 - 1) Equipment input data;
 - 2) Load flow study;
 - 3) Equipment Heat loss calculation;
 - 4) Cable sizing calculation;
 - 5) Voltage drop calculation;
 - 6) Motor starting study;
 - 7) Short-circuit study; and
 - 8) Protective device co-ordination study.

(c) Quality Assurance

- (i) Short-circuit, and protective device co-ordination shall be prepared in accordance with IEC 60909 by the Contractor's LEW.

(d) Sequencing and Scheduling

- (i) An initial, complete short-circuit study must be submitted and reviewed before S.O. will approve any electrical equipment, shop drawings for high and low voltage switchgears and transformers.

- (ii) Initial complete protective device co-ordination studies shall be submitted within 90 days after approval of initial short circuit study.
 - (iii) The revised short-circuit, and protective device co-ordination studies shall be submitted 30 days before energizing electrical equipment. Utilise characteristics of as-installed equipment and materials.
 - (iv) Final short circuit, and protective device co-ordination shall be completed prior to project issuance of Certificate of Substantial Completion. Final version of study shall include as-installed equipment, materials, and parameter data or settings entered into equipment based on study.
- (e) General
 - (i) Equipment and component titles used in the studies shall be identical to the equipment and component titles shown on the Drawings.
 - (ii) Perform studies using one of the following digital computer software packages:
 - 1) SKM Power Tools;
 - 2) ETAP; or
 - 3) Equivalent approved by S.O..
 - (iii) Perform complete load flow for each equipment in normal and worst case operating scenario.
 - (iv) Perform voltage drop calculation in accordance with CP 5 for normal and worst case operating scenario.
 - (v) Perform complete fault calculations for each (proposed and ultimate) source combination for different operating scenarios.
 - (vi) Source combination may include present and future power company supply circuits, large motors, or generators.
 - (vii) Utilise proposed load data for the study obtained from the specifications, drawings, wiring information and equipment.
- (f) Device co-ordination time-current curves for high and low voltage distribution system(s), including individual protective device time-current characteristics.
- (g) Individual protective device time-current characteristics on electronic media.

G5.50.2 Load Flow Study and Voltage Drop Calculations

- (a) Load flow study shall be carried out for the entire power system up to the end loads / final circuits.
- (b) All connected loads shall be modelled in the power system.
- (c) Load flow study shall be run prior to sizing of the transformers and cables. Load flow study results shall be shown on SLD and report comprising voltage parameters, power parameters including power factor.
- (d) Voltage drop calculations shall be carried out in accordance with CP 5. Voltage drop across each cable and at each bus and equipment shall be shown on SLD and report.
- (e) Voltage dip during motor starting shall be calculated and shall comply with SPPG requirements at Point of Common Coupling and maintain power system reliability.

G5.50.3 Short-Circuit Study

- (a) General
 - (i) Prepare in accordance with IEC 60909;
 - (ii) Use cable impedances based on copper conductors, except where aluminium conductors are specified or shown;
 - (iii) Use bus impedances based on copper bus bars, except where aluminium bus bars are specified or shown;
 - (iv) Use cable and bus resistances calculated at 30 degrees C;
 - (v) Use high voltage cable reactances based on actual data from manufacturer;
 - (vi) Use 600 volt low voltage cable reactances based on published data from the manufacturer or applicable tables in SS CP 5;
 - (vii) Use transformer impedances 92.5 % of “nominal” impedance based on tolerances specified in IEC;
- (viii) Submit
 - 1) Calculation methods and assumptions;
 - 2) Actual Calculations;
 - 3) Tabulation of calculated quantities. Results, conclusions and recommendations. Selected base per unit quantities;
 - 4) Separate Single-line diagrams for Input Data, Load Flow results, Voltage Drop results, and Short Circuit results;
 - 5) Source impedance data, including electric utility system and motor fault contribution characteristics;
 - 6) Impedance diagrams; and
 - 7) Zero sequence impedance diagrams.
- (ix) Calculate short-circuit interrupting and momentary (when applicable) duties for an assumed three-phase bolted fault at each:
 - 1) Electric utility’s supply termination point;
 - 2) Substation high voltage switchgear;
 - 3) Distribution high voltage switchgear;
 - 4) High and low voltage side terminals of all transformers;
 - 5) Low voltage switchgear and switchboards;
 - 6) Motor control centres;
 - 7) Standby and emergency generators;
 - 8) All distribution panel boards;
 - 9) All branch circuit panel boards;
 - 10) Other significant locations throughout the system; and

- 11) Future load contributions as shown on single-line diagrams or in specifications.
- (x) Provide short circuit study for single phase-to-ground fault current;
- (xi) Provide short circuit study for 3-phase bolted fault (L-L-L and L-L-L-G) and line-line fault;
- (xii) Verify
 - 1) Equipment and protective devices are applied within their ratings;
 - 2) Adequacy of switchgear, switchboard, distribution panel boards, and motor control centres bus bars to withstand short-circuit stresses;
 - 3) Adequacy of transformer windings to withstand short-circuit stresses; and
 - 4) Cable and busway sizes for ability to withstand short-circuit heating, besides normal load currents.
- (xiii) Tabulations
 - 1) General Data:
 - Short circuit reactances of rotating machines.
 - Cable and conduit material data.
 - Bus data.
 - Transformer data.
 - Circuit resistance and reactance values.
 - 2) Short Circuit Data [(for each source combination)]:
 - Fault impedances.
 - X / R ratios.
 - Asymmetry factors.
 - Motor contributions.
 - Short circuit kVA.
 - Symmetrical and asymmetrical fault currents.
 - 3) Equipment Evaluation:
 - Equipment bus bracing, equipment short circuit rating, transformer, cable, busway.
 - Maximum fault current available.
- (xiv) Written Summary
 - 1) Scope of studies performed;
 - 2) Explanation of bus and branch numbering system;
 - 3) Prevailing conditions;
 - 4) Selected equipment deficiencies;
 - 5) Results of load flow, voltage drop and voltage dip study;

- 6) Results of short circuit study; and
- 7) Comments or suggestions.
- (xv) Suggest and implement changes and additions to equipment rating and/or characteristics; and
- (xvi) Notify S.O. in writing of existing circuit protective devices improperly rated for fault conditions and rectify deficiencies.
 - 1) Revise data for “as-installed” condition.

G5.50.4 Protective Device Co-ordination Study

- (a) General
 - (i) Prepare in accordance with IEC 60909.
 - (ii) Proposed protective device co-ordination time-current curves for distribution system, graphically displayed on conventional log-log curve sheets:
 - 1) Provide separate curve sheets for phase and ground fault coordination for each scenario.
 - 2) Each curve sheet to have title and single-line diagram that applies to specific portion of system associated with time-current curves on that sheet. Limit number of devices to four to six.
 - 3) Terminate device characteristic curves at a point reflecting maximum symmetrical or asymmetrical fault current to which device is exposed.
 - 4) Identify device associated with each curve by manufacturer type, function, and, if applicable, tap, time delay and instantaneous settings recommended.
 - (iii) Plot Characteristics Curve
 - 1) Power Grid relays;
 - 2) High voltage equipment relays;
 - 3) High and low voltage circuit breakers including manufacturer’s minimum melt, total clearing, tolerance and damage bands;
 - 4) High and Low voltage equipment circuit breaker trip devices, including manufacturer’s tolerance bands;
 - 5) Pertinent transformer full-load currents at 100 %;
 - 6) Transformer magnetising inrush currents;
 - 7) Transformer damage curves; appropriate for system operation and location;
 - 8) Transformer generator and supply feeders damage curves;
 - 9) ANSI / IEC transformer withstands parameters;
 - 10) Significant symmetrical and asymmetrical fault currents;
 - 11) Ground fault protective device settings;

- 12) Other system loads protective devices for largest branch circuit and feeder circuit breaker in each motor control centre;
- 13) Motor starting curves; and
- 14) HV and LV cables damage curves.
- (iv) Primary Protective Device Settings all Transformers:
 - 1) Secondary Line-to-Ground Fault Protection: Primary protective device operating band within the transformer's characteristics curve, including a point equal to 58% withstand point.
 - 2) Secondary Line-To-Line Faults: 16 % current margin between primary protective device and associated secondary device characteristic curves.
- (v) Separate 415 V, 6.6 kV, and 22 kV relay characteristics curves from curves for other devices by at least 0.4-second time margin.
- (vi) Co-ordination curves for circuit breakers feeding equipment provided and installed in other contracts shall be prepared using information obtained from the contractor supplying the equipment being fed and shall show coordination with the next device downstream.
- (vii) Contractor shall obtain data of existing system and carry out coordination curves between existing system and new system.

G5.50.5 Tabulations

- (a) General Data
 - (i) Short-circuit reactances of rotating machines;
 - (ii) Cable and conduit material data;
 - (iii) Bus data;
 - (iv) Transformer data; and
 - (v) Circuit resistance and reactance values.
- (b) Short-Circuit Data
 - (i) Fault impedances;
 - (ii) X/R ratios;
 - (iii) Asymmetry factors;
 - (iv) Motor contributions;
 - (v) Short-circuit kVA; and
 - (vi) Symmetrical and asymmetrical fault currents.
- (c) Recommended Protective Device Settings
 - (i) Relays
 - 1) Current tap;
 - 2) Time dial; and
 - 3) Instantaneous pickup.

- (ii) Circuit Breakers
 - 1) Adjustable pickup;
 - 2) Adjustable time-current characteristic;
 - 3) Adjustable instantaneous pickup;
 - 4) All available protective functions;
 - 5) I^2t In/Out;
 - 6) Zone interlocking; and
 - 7) Electronic settings data file.

G5.50.6 Study Analysis

- (a) Written Summary
 - (i) Scope of studies performed;
 - (ii) Explanation of bus and branch numbering system;
 - (iii) Prevailing conditions;
 - (iv) Selected equipment deficiencies;
 - (v) Results of short-circuit and co-ordination studies;
 - (vi) Appendix of complete relay and circuit breaker electronic setting files and submit electronic data files from manufacturer's software; and
 - (vii) Comments or suggestions.
- (b) Suggest changes and additions to equipment rating and/or characteristics.

G5.50.7 Execution

- (a) General
 - (i) Adjust relay and protective device settings according to values established by co-ordination study.
 - (ii) Make minor modifications to equipment as required to accomplish conformance with the short-circuit and protective device co-ordination studies.
 - (iii) Notify S.O. in writing of any required major equipment modifications.

END OF SECTION